

SMART INDIA HACKATHON 2026



RASTA AI

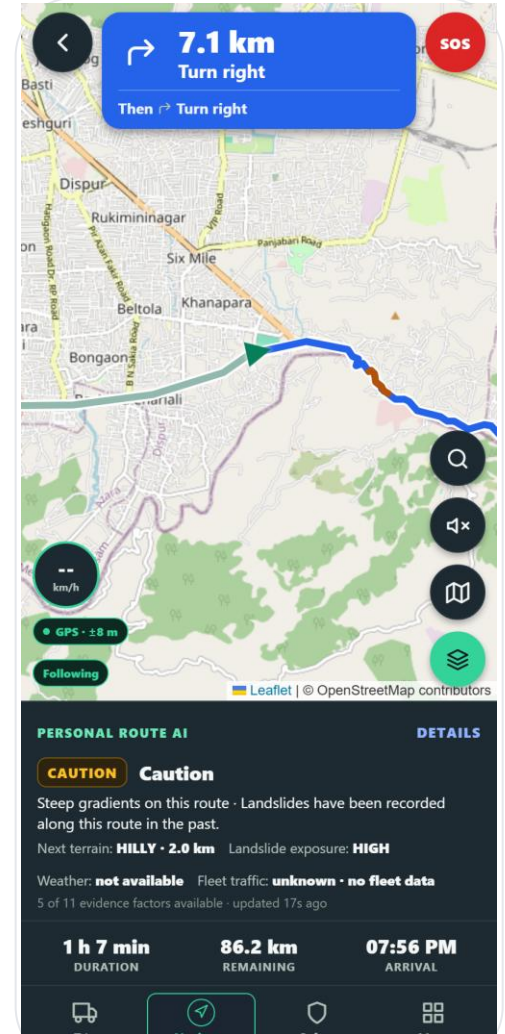
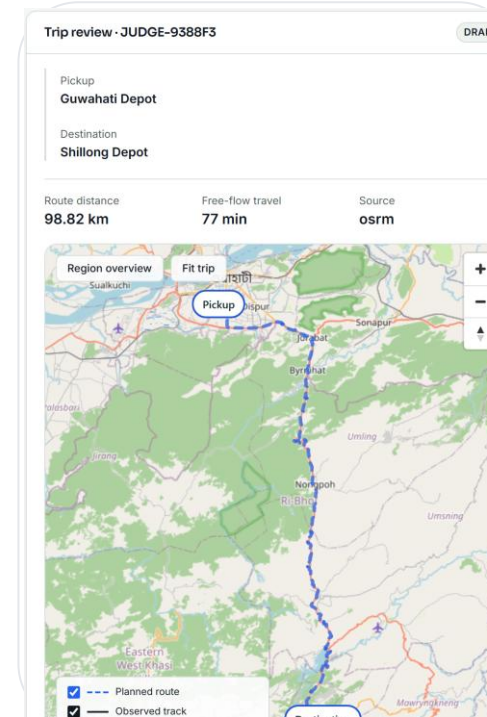
Route Intelligence for Essential-Supply Logistics in North Eastern India

SIH26002

Problem Statement ID
MDoNER · Smart Automation · Software

PROBLEM STATEMENT ID	SIH26002
PROBLEM STATEMENT TITLE	AI-Based Smart Logistics and Accessibility Intelligence Platform for North Eastern Region (NER)
THEME	Smart Automation
PS CATEGORY	Software
TEAM ID	17 (internal group no.)
TEAM NAME	NER-AI LOGISTICS

End-to-end working prototype · physical-phone validated · live demo corridor Guwahati → Shillong



Manager route review · Driver navigation — real screens

PROPOSED SOLUTION

NOT THE SHORTEST ROAD. THE ROAD THAT IS OPERATIONALLY USABLE NOW.

THE PROBLEM ON NER CORRIDORS

- Mountain terrain
- Monsoon weather
- Landslides
- Flood context
- Road disruptions
- Weak connectivity
- Missing or stale evidence

A shortest-path router still sends the truck.

WHAT IS DIFFERENT

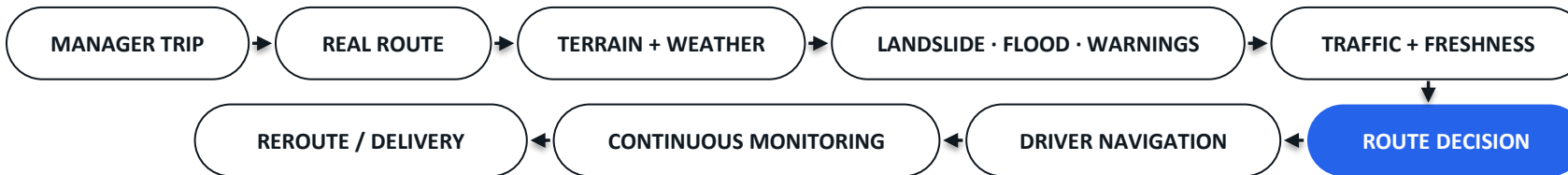
Evidence-aware routing — terrain, weather, warnings, flood, landslide history, traffic, freshness; not shortest-path only

Unknown or stale data is exposed — never treated as safe

Human-governed rerouting — a manager approves every safety-critical route change

23 languages + offline operation — built for drivers on NER corridors

THE RASTA FLOW



UNKNOWN ≠ SAFE

Missing or stale evidence is shown as UNKNOWN and never counted as safe. Routes with unknown hazard data need an authorised reviewer before selection.

REQUIRES REVIEW

Hazard evidence is incomplete or elevated. An authorised reviewer must review this route before selection. Refresh conditions after review.

Weather data not available · Weather readings are old · Landslide data is not available · no source is connected. · Steep gradients on this route · Landslides have been recorded along this route in the past. · The landslide inventory ends in 2017 - recent years are not covered. · River levels near this route are close to their recent average. · Official disaster alerts could not be checked for this route. · No fleet truck has driven this road in the last fifteen minutes, so traffic is unknown.

Unavailable: Fuel range, Landslide, Official alerts, Road quality, Truck restrictions, Weather

Evidence coverage · Weather UNKNOWN · Terrain AVAILABLE · Warnings UNKNOWN · Traffic UNKNOWN

TERRAIN

0.5 KM STEEP

49–1436 m · 1975 m climb · steepest 10.3%

flat 61.8 km · rolling 29.6 km · hilly 7.0 km · steep 0.5 km

Copernicus DEM GLO-90 via Open-Meteo + SRTM 30 m via OpenTopoData · 100% of 198 samples answered

LANDSLIDE HISTORY

HIGH EXPOSURE

22 recorded landslides within 5 km of the road, nearest 0.4 km

6 more nearby placed too imprecisely to count · HIGH is three or more; MODERATE is one or two

NASA Global Landslide Catalog · inventory 2007–2017 — aged, recent years not covered

RIVER LEVELS

NORMAL

Discharge at 5 river cells along the corridor, highest 1.0× its own 30-day mean · GloFAS via Open-Meteo, 2026-09-14. A level, not a flood forecast and not a road-closure claim.

OFFICIAL ALERTS

NOT CHECKED

The alert feed or the district lookup did not answer for this corridor.

FLEET TRAFFIC

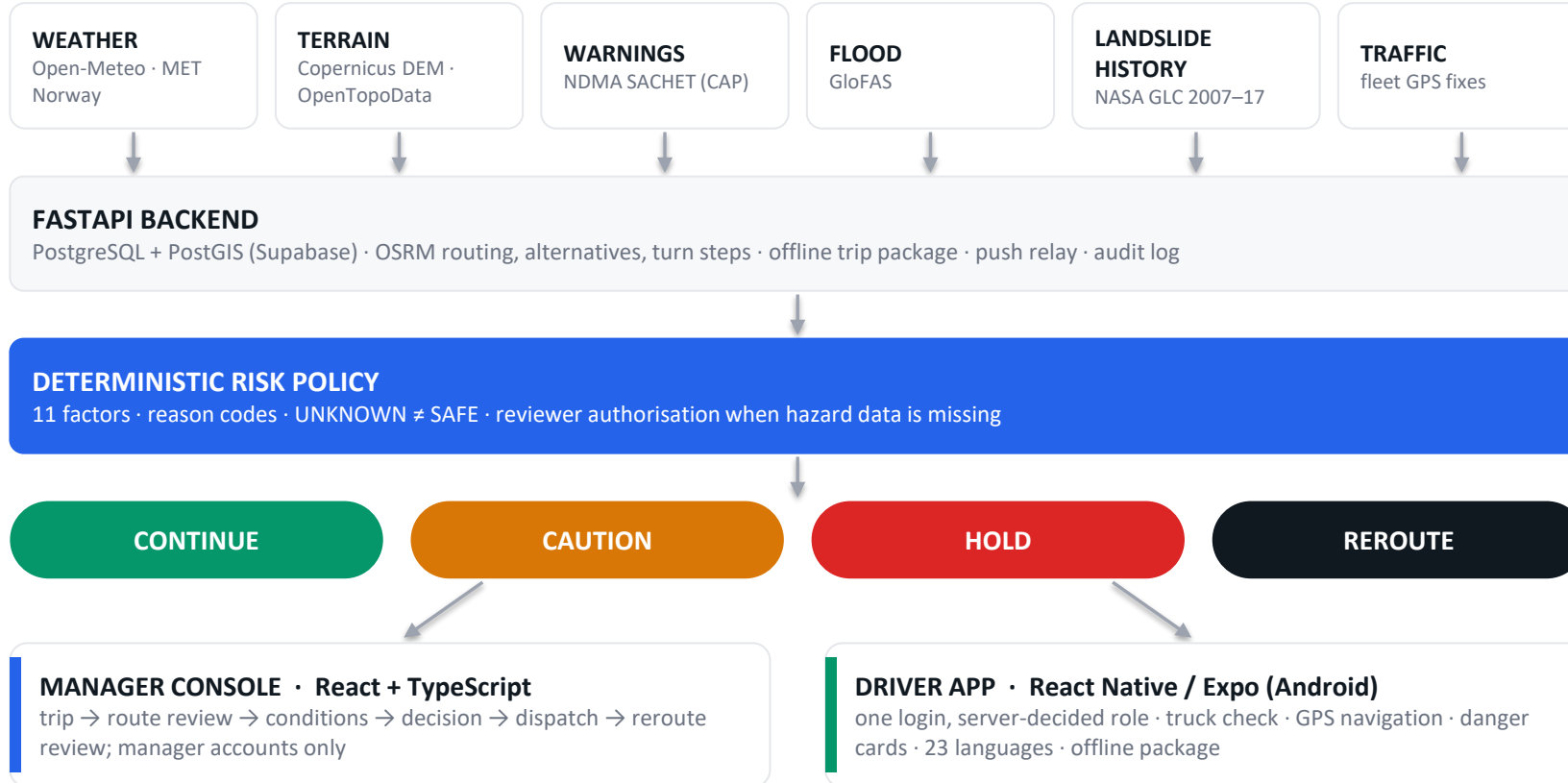
UNKNOWN

No RASTA truck has driven this road in the last 15 minutes. Unknown, not

Manager · Check conditions (real screen)

TECHNICAL APPROACH

VERIFIED DATA · provider adapters with health + freshness



STACK FastAPI · PostgreSQL / PostGIS (Supabase) · React · React Native / Expo · OSRM · OpenStreetMap · Gemini

EVIDENCE Open-Meteo / MET Norway · NDMA SACHET · GloFAS · NASA historical landslides · OpenTopoData / Copernicus DEM

METHOD plan → score evidence → decide → dispatch → track (60 s route-ahead worker) → reassess → human reroute → deliver

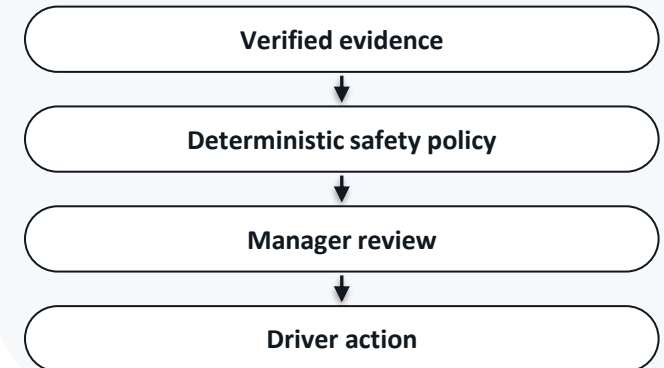
AI ARCHITECTURE — WHO DECIDES

DETERMINISTIC SAFETY ENGINE
makes every operational decision

AI / LLM (Gemini, OpenRouter fallback)
explains and assists in the driver's language — never decides

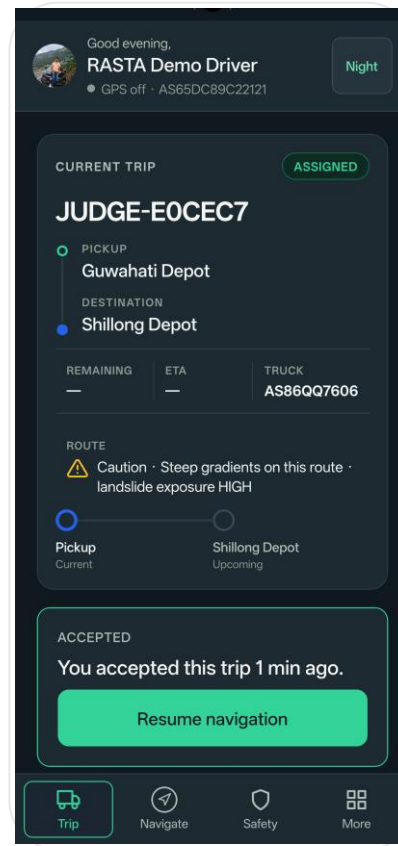
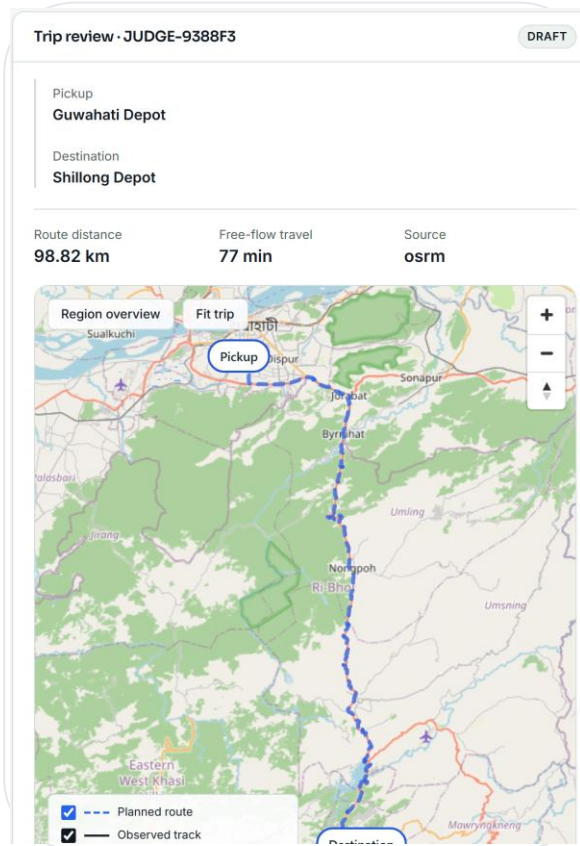
EXPERIMENTAL ML
research channel only; held out of routing by a validation gate

GOVERNANCE CHAIN



NOT A CONCEPT — AN END-TO-END WORKING PROTOTYPE

Hosted backend, manager console and the Android app on a physical phone, on real roads.



Real screens · manager route review · driver trip

WORKING TODAY

- ✓ Manager login (manager-only console)
- ✓ Driver login · role-aware mobile UI
- ✓ Real OSRM route + alternatives
- ✓ Terrain, weather, flood, landslide evidence
- ✓ Truck photo verification
- ✓ Physical GPS navigation
- ✓ Off-route detection
- ✓ Manager-approved reroute
- ✓ Offline / degraded operation
- ✓ Stops → delivery → fleet available

KNOWN LIMITS → MITIGATION

Render cold start → warm /health before the demo

Provider rate limits → cached evidence + explicit UNKNOWN

VALIDATION

1138+

backend tests

621+

driver-app tests

170+

manager-web tests

PHYSICAL ANDROID · CERTIFIED

Canonical demo Guwahati → Shillong
 ≈ 98.8 km real route · 9/9 role steps · 12/12 judge-flow steps on the phone

Push needs Firebase config → in-app safety alerts already work

Experimental ML → outside routing until the gate passes

IMPACT AND BENEFITS

RASTA AI does not ask only “Which road is shortest?” It asks “Can this truck reliably use this corridor now?”

WHO BENEFITS

FLEET MANAGERS

Route usability and uncertainty visible before dispatch; auditable decisions.

DRIVERS

Safer navigation, danger context, emergency support, 23 languages.

REMOTE COMMUNITIES

More resilient access to food, medicine, fuel and relief supplies.

GOVERNMENT / OPERATIONS

Auditable route evidence and disruption visibility per corridor.

DEMONSTRATED TODAY

23 selectable languages, explicit fallback status

12/12 judge-flow steps certified on a physical phone

11 factor route-risk assessment

100 % of reroutes reviewed by a human

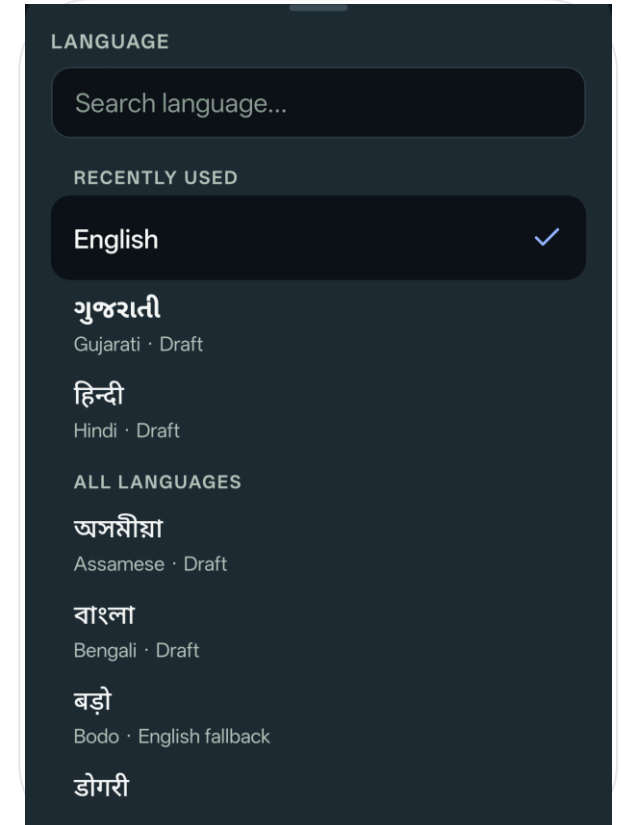
60 s route-ahead coordinator

0 unknown factors counted as safe

HUMAN GOVERNANCE — AI HAS NO UNCONTROLLED AUTHORITY



Impact is framed on essential logistics — food, medicine, fuel, relief — and on evidence, not on unmeasured savings.



Driver · language selector (real screen)

23 languages in the selector; each shows its status — Verified, Draft or English fallback — so a driver is never misled.

RESEARCH / EVIDENCE SOURCES

NASA Global Landslide Catalog

historical landslide exposure; inventory 2007–2017, ≤5 km accuracy filter

NDMA SACHET (CAP feed)

official warnings, polled with freshness

GloFAS — Copernicus Emergency Management

flood context: river discharge vs 30-day mean

Open-Meteo · MET Norway

weather forecast along the corridor; ERA5-Land archive for research

Copernicus DEM GLO-90 · OpenTopoData

terrain profile, climb, steep segments

OpenStreetMap · OSRM

roads, route alternatives, turn steps

MDoNER problem statement SIH26002

AI-based smart logistics and accessibility intelligence for the NER

MODEL VALIDATION — EXPERIMENTAL LANDSLIDE MODEL

Logistic regression on NASA GLC events + ERA5-Land rain + DEM slope. India-wide training (605 events, 3,025 rows) · NER geographic holdout (770 days, 154 events) · temporal split ≤2014 / 2015–17 · threshold chosen on train only.

0.95

Recall

0.32

Precision

0.55

PR-AUC

0.83

ROC-AUC

0.50

FPR

EXPERIMENTAL — NOT DEPLOYED

rule baseline: recall 0.25 @ FPR 0.03

Validation before automation. Experimental ML stays outside production until it meets the required geographic reliability and false-positive thresholds.

The false-positive floor is a data-design limit (same-site quiet-day negatives; false positives are wet monsoon days), measured in a full error analysis. Next data: sub-daily rain intensity, soil moisture, cross-site negatives.

REFERENCES

gpm.nasa.gov/landslides (NASA GLC) · sachet.ndma.gov.in · global-flood.emergency.copernicus.eu (GloFAS) · open-meteo.com · api.met.no · opentopodata.org · Copernicus DEM · project-osrm.org · openstreetmap.org · github.com/nxtlucifer/ner-ai-logistics (repository, tests, model registry, error analysis)